Experiment 2: Cosmic Muons

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October 18th 2024

1 Abstract

The main objective of the experiment is to determine the angular distribution of muon cosmic rays. Cosmic rays, also known as astroparticles are particles that move through space at the speed of light. Stable nuclei with longer lifetimes can be found when cosmic radiation incident at the very top of the terrestrial atmosphere. Primary cosmic rays, high energy particles like protons, electrons that typically originate from astrophysical sources and bombards the earths atmosphere which results in the creation of secondary cosmic rays, due to the collision of high energy particles with the atoms in the atmosphere. Secondary particles are Lithium, Beryllium, pions and Boron. Muons are produced when pions decay at an altitude of 15km. At the Edwards Laboratory we were able to observe the nature of muons for the varying zenith angle.
Word count: 133

2 Background Introduction

Muons are one of fundamental subatomic particles that can be found in the universe. They move at a speed of light and therefore travels far before decay. The lifetime of the muon is considered extremely important as it validates the Einsteins theory of Special Theory. This experiment is conducted to determine the rate of muons with the use of the scintillator telescope. When charged particles passes through transparent materials there's a possibility of ionization or excitation resulting in a flash when it de-excites. This is known to be scintillation. These flashes are detected by Photomultiplier tubes (PMT) that are extremely light sensitive and it converts the flashes of light to identifiable pulses. The primary objective of this experiment is to identify that the zenith angle relation with the rate of muons detected is in fact a square cosine relation. Scintillator at the Edwards laboratory was used in this task. Therefore the main task is to identify the count rates. The threshold voltage is the voltage that necessarily would give the count rates of muons as if the voltage is higher than the threshold voltage we will be counting other noise related particles as well. The threshold voltage was given to the four discriminators. Afterwards the rate of the muons with the respect to the zenith angle was obtained to show that it has a square cosine relation. The piDAQ was used to analyze the delay spectrum in order to obtain the time calibration graph. This was done to calibrate the TAC: Time to Analog converter.
Word count: 255

3 Theory

Cosmic radiation bombards the atmosphere from outside the solar system. The energy of the cosmic rays is around 1GeV to 10^8 TeV. The rate in which the muons strikes the atmosphere decreases with increasing energy [1]. There's an anti-correlation between the solar activity and the intensity of gamma rays lower

than 10GeV. These cosmic rays are affected by the geomagnetic field of the earth which is required when penetrating the top of the atmosphere. Therefore the low energy cosmic radiation depends on the location and the time.

The rate of cosmic particles decreases when the zenith angle increases. When the zenith angle is at 90 deg the rate of cosmic particles is at its minimal value. Relationship between the zenith angle and the count rate of the muons shows a cos^2 function.
Muons are produced due to decay of pions at an altitude of 15 Km and the known to be unstable with a lifetime of 2.2 μs.[2] Due to special theory of relativity, the consideration of time dilation as the muon travels at a speed approximately equal to the speed of light needs to be taken into value when calculating the distance traveled. Therefore at the angle of 90 deg the muon travels a distance of 400 Km through the atmosphere to the surface of the earth.[2] This can be done by high energy muons.

As a result of conservation of momentum muons travels in the same direction as their parent pions. When the pions interacts with the atmosphere head on, it reaches the more denser lower atmosphere resulting in the collision of particles which leads to forming of more pions. If the energy of the pions are low that there's high probability of decay than interaction with other particles, this leads to the formation of muons. Muons are also formed when pions reaches the earth at a higher angle of incidence and any other interactions are less likely to take place. A charged particles that has no interaction decays into a muon and a neutrino after 0.028 μs. [2] The muon is known as an unstable lepton particle, charge (-e) that decays into an electron, neutrino and an anti-neutrino. Anti-muon also known as the positive muon has a positive charge and similarly decays into a positron, neutrino and an anti-neutrino.

The kinetic energy of the charged particles the muons and the decay electrons deposits energy in the scintillator material due to the Coulomb interaction. Some of the deposited energy will produce a flash of light. And a second flash of light will be produced if the muon decays inside the scintillator. By measuring the time distribution between two flashes for large ensemble of particles it is possible to determine the mean lifetime of the muon. If they decays whilst moving in the scintillator then the two flashes will be seen as one continuous flash.
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4 Experimental Methods

At the Edwards Laboratory, Ohio University we are using a scintillator telescope which is made of two scintillators that are separated by a lever arm to determine the angular distribution of cosmic muons rays.

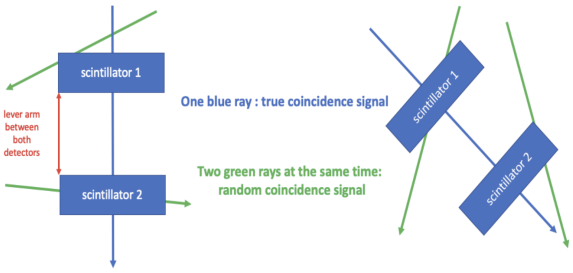


Figure 1: Scintillator Telescope [2]

The telescope can be rotated to obtain the mean direction in which the muons are passing through both detectors. As seen in Figure 1 the blue ray passes through both the detectors simultaneously producing the signals in coincidence in a defined direction whereas the green ray passes through both detectors simultaneously at random coincidence. Figure 2 shows the electronic chain that measures the muon lifetime. The photo-multiplier tube (PMT) reads each scintillator in the left and the right. A logic signal is produced when

the PMT analog signal is higher than the threshold of the discriminator. If two logic signals are produced at the output simultaneously then the coincidence unit produces an output signal. The Time to Analog converter (TAC) is used to produce a logic signal where the amplitude is proportional to the time in between the start and the stop signals.

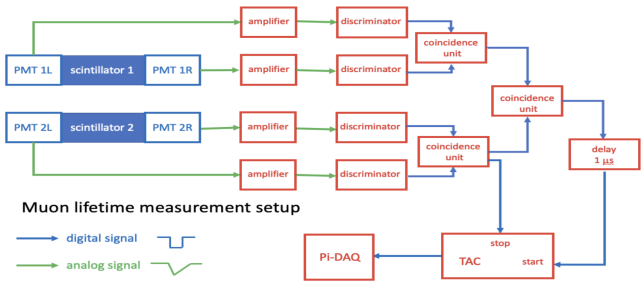


Figure 2: Electronic chain [2]

Discriminator: If the amplitude of the input analog signal is larger than the threshold the discriminator outputs a digital signal. A potentiometer is used to adjust the threshold voltage. There are two PMTs paired for each discriminator. The operating voltage for the PMTs is 1320V [2].

TAC: This gives out the output voltage which is proportional to the time difference between the start and stop signals. Output is given as a histogram. The calibration of the horizontal axis is done by giving the TAC separations that are in-between the start and the stop that can be controlled. Therefore the time delay can be found for any specific channel of the histogram.

4.1 Procedure

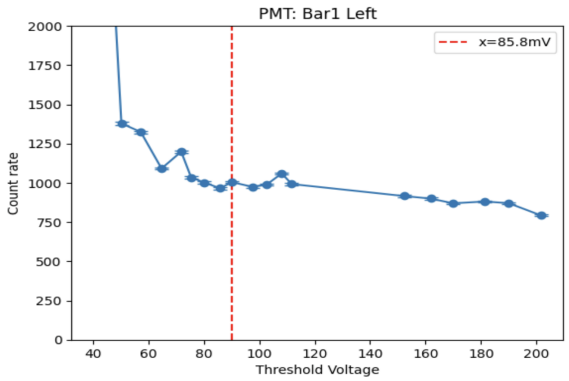
The first task was to identify the threshold voltage. This was done by getting the number of counts for a constant period of time in this case, 20 seconds for a certain voltage. The range of voltage we chose was around 40 mV to 200 mV. Main goal was to identify the plateau region. The voltage corresponding to the plateau region is considered to be the threshold voltage. This was done to tune out any events that aren't occurred by muons. The coincidence units were used to determine the presence of muons. It was observed that the time taken for an event to happen was significantly lesser in between two scintillators in comparison to each coincidence unit. The time duration for the separate coincidences and the double coincidence is 30 seconds each. The next objective was to determine the angular distribution variation, the relationship between the zenith angle and the rate of muons detected. This was conducted my rotating the telescope starting from 90 deg to the ground level. A blob was connected at the center aligning to a protector to determine the corresponding angles. The final part of the experiment was the Time calibration. The TAC was used to generate the voltage signal corresponding to the given delayed pulse. The range of the delayed pulses used 1-8 μs.

4.2 Observations

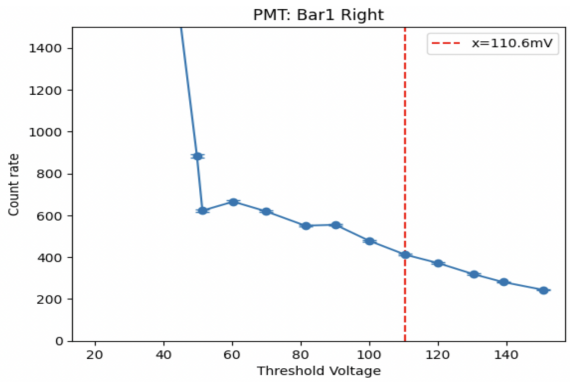
Determining the threshold voltage was bit complex as seen in Figure 3 the plateau region wasn't exactly definite. The main point was to find the voltage where the count rates showed lesser variation. Afterwards we got the count rate for each PMTs left and right separately and the count rate for the coincidence units. It was observed that the count rate was lesser than prior. The double coincidence has much lower count rate compared to the other which meant that not all events occurred in the scintillator are muon events and there's probabiltiy for some muons to have not passed through the scintillators.
The error of the count rates was found using Poisson statistics where the error has a dependency on $\sqrt{N}$.
Word count: 631

5 Analysis and Results

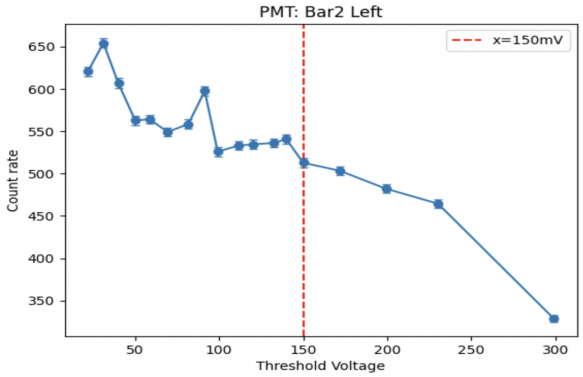
The Figure 3 shows the detection of the threshold voltages. It is visible that the voltage that has a few variations in the count rate is taken as the threshold voltage.



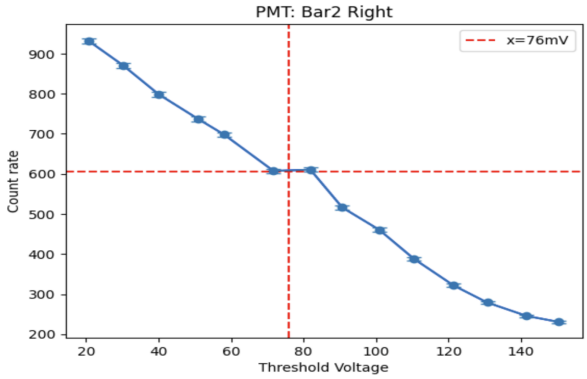
(a) PMT-1 Left



(b) PMT-1 Right



(c) PMT-2 Left



(d) PMT-2 Right

Figure 3: Bias voltage: t = 20 sec

Table 1 shows the obtained threshold voltage and the corresponding count rates for 30 seconds each.

Table 1: Individual counts

PMT	Threshold voltage (mV)	counts	count rate (counts/sec)	Error (count rate)
Bar1 left	85.8	32440	1081.33	6.00
Bar1 right	110.6	9514	317.13	3.26
Bar2 left	150	14524	484.13	4.02
Bar 2 right	76	16704	556.80	4.31

It can be observed that in Table 2 and 3, the two fold count rate for day 1 and day 2 have slight variations and in Table 4 the count rate has decreased, this is because all events aren't muon events.

Table 2: 2-fold count rate: Day 1

PMT Tube	**Day 1**		
	T1:Counts	**T1:Count-rates**	**T1:Error**
Coincidence 1	2942	98.07	1.81
Coincidence 2	3445	114.83	1.96

Table 3: 2-fold count rate: Day 2

PMT Tube	**Day 2**					
	T1:Count	**T1:Count-rate**	**T1:Error**	**T2:Count**	**T2:Count-Rate**	**T2:Error**
Coincidence1	2525	84.17	1.68	2567	85.57	1.69
Coincidence2	3250	108.33	1.90	3254	108.47	1.90

Table 4: 4-fold count rate: Day 1

T1:Counts	**T1:Count-rates**	**T1:Error**
139	4.633	0.392

Table 5: 4-fold count rate: Day 2

Trial 1			**Trial 2**			**Trial 3**			**Trial 4**		
Count	**Rate**	**Error**	**Count**	**Rate**	**Error**	**Counts**	**Rate**	**Error**	**Count**	**Rate**	**Error**
158	5.27	0.42	161	5.37	0.42	141	4.7	0.40	149	4.97	0.41

The angular distributions and the corresponding count rates are seen in Table 6. It can be seen that as the zenith angle increases the count rate decreases. This might be due to the fact that only high energy muons are able to reach when zenith angle is at 90 deg. The error of the angle reading is around 1 deg.

Table 6: Angle distributions and the corresponding count rates: t = 30 secs

θ **(deg)**	θ **(rad)**	**counts**	**count rate**	**Error (count rate)**
0	0.00	106	3.53	0.34
10	0.17	135	4.50	0.39
20	0.35	115	3.83	0.36
30	0.52	87	2.90	0.31
40	0.70	59	1.97	0.26
50	0.87	53	1.77	0.24
60	1.05	33	1.10	0.19
70	1.22	17	0.57	0.13
80	1.40	12	0.40	0.12
90	1.57	11	0.37	0.11

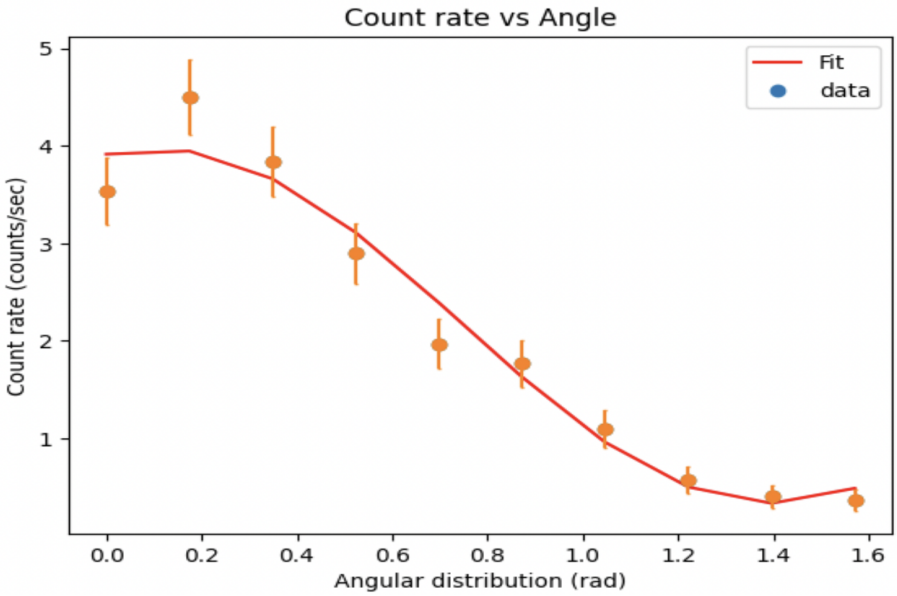


Figure 4: Square cosine dependence

Figure 4 represents count rate vs angle. Expression for count rate vs angular distribution:

$$f = A * (cos(w\theta - \phi))^2 + B$$

Table 7: Coefficients of Function fit values

Coefficient	**Magnitude**	**Error**
A	3.64	0.30
w	1.21	0.17
ϕ	-0.13	0.16
B	0.33	0.20

This shows that the experimental data values obtained align with the expected output that the count rate should decrease with the rise of the zenith angle and that the relationship should be a cos^2 relationship.

Figure 5 has a linear relationship between the count rate and the square of the cosine angle. The expression can be written as,

$$f = m * (cos(w\theta + \phi))^2 + c$$

Table 8: Coefficients of Function fit values

Coefficient	**Magnitude**	**Error**
m	3.64	0.25
c	0.33	0.13

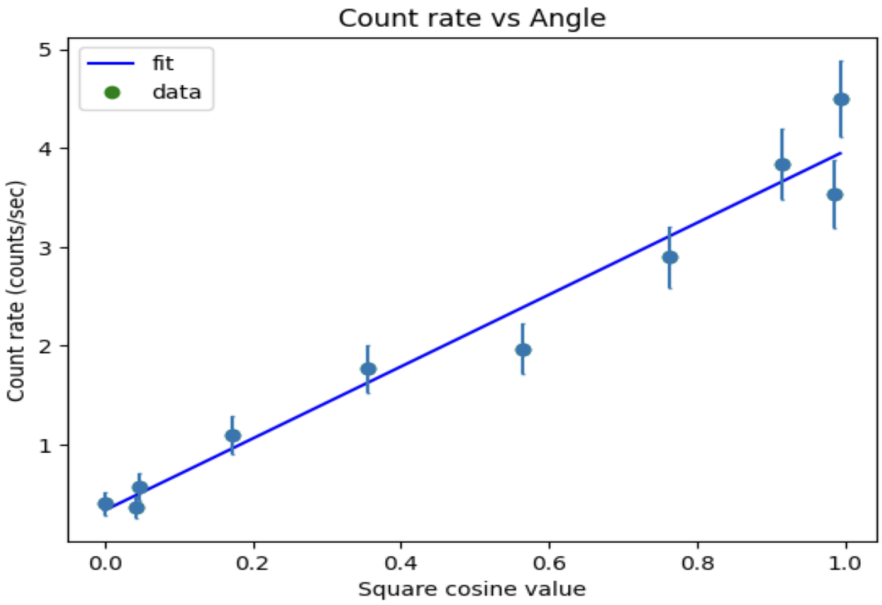


Figure 5: Linear Relation

Table 9: Residual of the Count rates

Experimental	**Functional**	**Residual**
3.53	3.91	0.38
4.50	3.95	0.55
3.83	3.66	0.17
2.90	3.11	0.21
1.97	2.39	0.42
1.77	1.63	0.14
1.10	0.96	0.14
0.57	0.50	0.07
0.40	0.33	0.07
0.37	0.49	0.49

The residual of the count rates can be seen in Table 9. It can be observed that the difference is comparatively small and has a range of 0.07-0.5.

Table 9 shows the calibration of the TAC to ensure the accuracy of the measurements which are important when calculating the muon decay time. The setup was given a delayed pulse and then the time to amplitude converter gives the data as a voltage signal. This spectrum was graphed in DAQ that plots the counts vs the channel number as a histogram as seen in Figure 6. Time duration for each delay was approximately one minute. Each of these peaks were then evaluated to obtain the mean bin, rms, total count and error. It can be assumed that the error in measuring time can be about 1/10th of a microsecond.
A linear regression fit can be obtained using the mean bin number calculated above to the corresponding delay time. The calibration line can then be using to obtain the time decay for other peaks whenever the particles are detected.

Table 10: Delayed timing events

Peak	Time μs	Total count	Mean (bin)	RMS (bin)	Error
1	0.44	240.00	1480.03	0.67	0.04
2	1.58	313.00	2645.00	2.18	0.12
3	2.80	263.00	3805.00	5.16	0.32
4	4.00	251.00	4959.00	7.44	0.47
5	5.14	251.00	6071.00	8.17	0.52
6	6.30	165.00	6244.00	8.37	0.65
7	6.98	309.00	6767.00	8.15	0.46

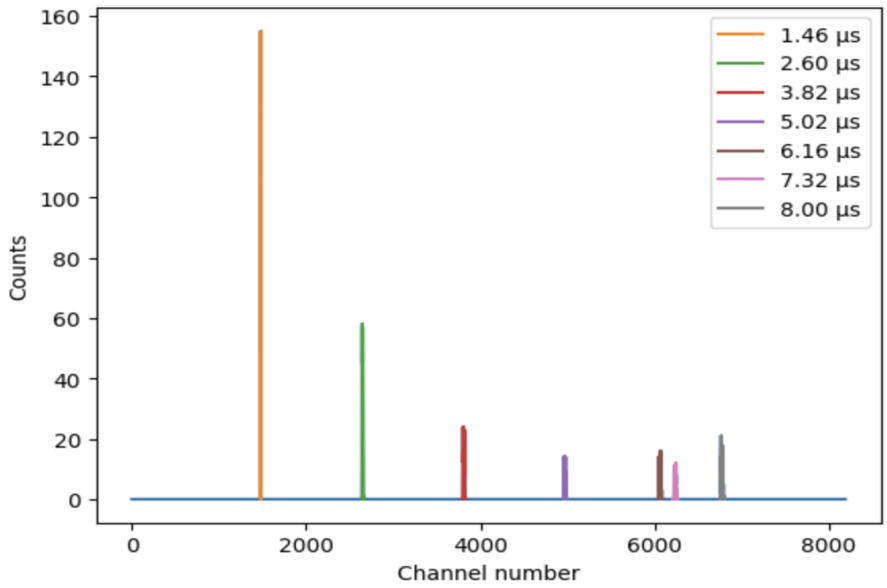
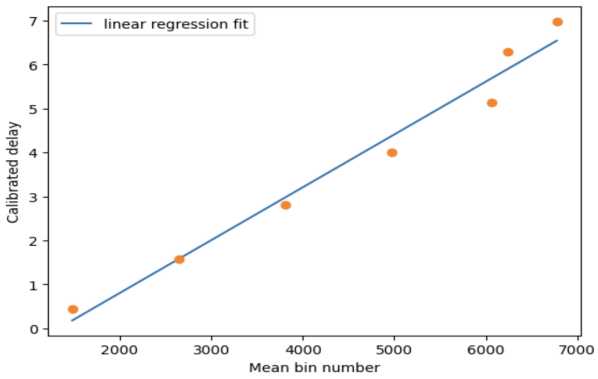
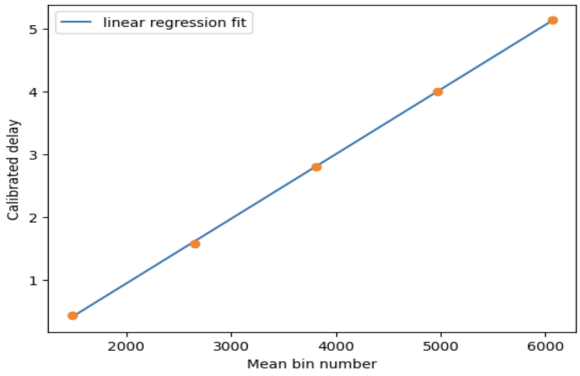


Figure 6: Delay time events

If the last two data points in Figure 7(a) are excluded it can be observed that the function fit will be in total agreement with the experimental data as seen in Figure 7(b). That would mean that there might have been errors during the collection of data.



(a) Linear regression fit



(b) Linear regression fit excluding the last two points

Figure 7: Linear Regression

Table 11: Residuals of the delay calibration

Experimental	**Figure 7(a)**		**Figure 7(b)**	
	Functional	**Residual**	**Functional**	**Residual**
0.44	0.18	0.27	0.43	0.01
1.58	1.58	0.00	1.61	0.03
2.80	2.98	0.18	2.80	0.01
4.00	4.37	0.37	3.97	0.03
5.14	5.69	0.54	5.16	0.02
6.30	5.90	0.40		
6.98	6.54	0.44		

Table 10 demonstrates the residual delay times and it can be observed that the difference between the experimental and functional values are in the range of 0-0.55 when all data values are taken into consideration but when the last two values are excluded the residual is in the range of 0.01 to 0.03 which can be significantly low elaborating the accuracy of the data.
There were some variations when choosing the threshold voltage. As it can be observed the variation in the count rate is extremely small for several voltage values. Therefore the same experiment was done with both sets of data values. The experimental data values does not show a larger variation since the difference in the count rates were small. This report has been conducted based on dataset A.

Table 12: Comparing threshold voltages

Dataset	**Bar1 Left**	**Rate**	**Bar1 Right**	**Rate**	**Bar2 Left**	**Rate**	**Bar2 Right**	**Rate**
A	85.8	1081.33	110.6	317.13	150	484.13	76	556.80
B	97.5	1014	85.8	512	76.8	578	120.5	592

It can be seen in Table 12 that the count rate is higher in dataset B Bar 1 Right and Bar 2 Left but the rate is higher for Bar 1 Left and Bar 2 Right in dataset A. Even still the variations of the count rates aren't that high when looking at the big picture. Therefore it was observed that when examining the data in regard to the coincidence units the difference was extremely minute.
Word count: 638

6 Conclusion

The main key point to observe was the square cosine relation with the rate of the muons. The analysis shows that the higher coefficient represents the background event rate. Afterwards considering the square cosine to be the independent value the function of the count rate was observed. This shows a linear relationship. The final task was performing the linear time calibration to obtain the decay peaks. This detection leads to the fact that a moun was decayed inside the scinitillator resulting in the production of an electron which could be used in the half-life calculation. It was observed that in the delay calibration plot the last two data points cause a high residual error this might be due to the fact that the time duration for each delay was approximate and that might results in the variation of the number of counts. It was understood that a slight variation in choosing the threshold voltage has little to no impact on the analysis as the variation in the count rates are considerably small.

7 References

1 Cosmic rays: particles from outer space. (2024, October 17). CERN. https://www.home.cern/science/physics/cosmic-rays-particles-outer-space.

2 G.Jensen, P.M. King, and J. Roche, PHYS6751: Nuclear Graduate Lab manual v24.5, Ohio University, 2024